Hints: Probability – 2. Geometric probability and random points

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Fix the orientation $\theta \in [0, \pi/2)$; the position is then uniform modulo $\mathbb{Z}^2$. Let N be the number of lattice points covered. Then $\mathrm{E}N = \text{area} = 1$, so $\Pr(N = 0) = \sum_{k \geqslant 2} (k-1) \Pr(N = k)$.

Hint 2. A unit square (diameter $\sqrt{2}$) covers at most two lattice points up to a null set, and two covered points are adjacent. So $\Pr(N = 0) = \Pr(N = 2) = 2A(\theta)$, where $A(\theta)$ is the area of the set of positions covering both $(0,0)$ and $(1,0)$, i.e. the area of $S \cap (S + e_1)$.

Hint 3. In the square's own coordinates, $e_1 = (\cos\theta, -\sin\theta)$, so $A(\theta) = (1 - \cos\theta)(1 - \sin\theta)$. Average $2(1 - \cos\theta)(1 - \sin\theta)$ over $[0, \pi/2]$: the answer is $2 - \frac{6}{\pi}$.

Problem 2

Hint 1. The distance is $d = \min(x, 1-x, y, 1-y, z, 1-z)$; use $\mathrm{E}d = \int_0^\infty \Pr(d > t)\, dt$.

Hint 2. $\Pr(d > t) = (1 - 2t)^3$ for $0 \leqslant t \leqslant \frac{1}{2}$, so $\mathrm{E}d = \int_0^{1/2} (1-2t)^3\, dt = \frac{1}{8}$.

Problem 3

Hint 1. Let $g_1, \ldots, g_n$ be the arcs between consecutive points (circumference 1); they are uniform on the simplex. The angle at P_i is inscribed in the arc not containing P_i, so it is acute iff the two arcs adjacent to P_i have total length $> \frac{1}{2}$.

Hint 2. Two adjacent arcs have total length with a $\mathrm{Beta}(2, n-2)$ law, which gives $\Pr(\text{acute at } P_i) = n\, 2^{1-n}$. Two acute angles can only occur at adjacent vertices (two disjoint pairs of arcs cannot both exceed $\frac{1}{2}$), and never three.

Hint 3. Inclusion–exclusion: $\Pr = n \cdot n 2^{1-n} - n \Pr(g_{i-1} + g_i > \frac{1}{2},\ g_i + g_{i+1} > \frac{1}{2})$, and a direct integration over the simplex gives 2^{2-n} for the last probability. The answer is $\frac{n(n-2)}{2^{n-1}}$ (it equals 1 for $n = 4$, as it must for an inscribed quadrilateral).

Problem 4

Hint 1. Measure arcs by central angle. Find the arc C_1C_2: a point of the circle at central angle φ from O is at chord distance $2R \sin\frac{\varphi}{2}$, so C_1C_2 is the arc centred at O of half-angle $\varphi_0 = 2 \arcsin\frac{\pi}{4}$. Its complement is an arc of size $\beta = 2\pi - 4\arcsin\frac{\pi}{4} < \pi$.

Hint 2. The shorter arc AB has length L uniform on $[0, \pi]$ and a uniformly distributed position, independent of O. Given AB, the arcs do not meet iff O lies in an arc of length $(\beta - L)_+$, so $\Pr = \frac{1}{\pi} \int_0^\pi \frac{(\beta - L)_+}{2\pi}\, dL = \frac{\beta^2}{4\pi^2}$.

Hint 3. Thus $\Pr = \left(1 - \frac{2}{\pi}\arcsin\frac{\pi}{4}\right)^2 \approx 0.181$. (If r were measured along the circle, C_1C_2 would be a semicircle, $\beta = \pi$ and the answer $\frac{1}{4}$.)

Problem 5

Hint 1. The volume of the ball of radius $\sqrt{2}$ intersected with the unit cube: start from the octant of the ball, volume $\frac{\sqrt{2}\pi}{3}$, and subtract the parts where some coordinate exceeds 1 (inclusion–exclusion; two coordinates cannot both exceed 1 inside the ball).

Hint 2. The part with $x > 1$ is a quarter of a spherical cap of height $h = \sqrt{2} - 1$, volume $\frac{1}{4} \cdot \frac{\pi h^2(3\sqrt{2}-h)}{3} = \frac{(4\sqrt{2}-5)\pi}{12}$. So the volume is $\frac{\sqrt{2}\pi}{3} - \frac{(4\sqrt{2}-5)\pi}{4} = \frac{(15-8\sqrt{2})\pi}{12}$.

Hint 3. Now compute the complement in cylindrical coordinates over the unit square, using symmetry in $x \leftrightarrow y$: $1 - \frac{(15-8\sqrt{2})\pi}{12} = 2\int_0^{\pi/4} \int_1^{\sec\theta} (1 - \sqrt{2-\rho^2})\rho \, d\rho \, d\theta$. The inner integral equals $\frac{\tan^2\theta}{2} - \frac{1}{3} + \frac{1}{3}\big(2 - \sec^2\theta\big)^{3/2}$ and $(2 - \sec^2\theta)^{3/2} = \frac{\cos^{3/2} 2\theta}{\cos^3\theta}$; solve for the required integral.

Problem 6

Hint 1. Put $B = 0$, $A = -e$, $C = e$ with $|e| = 1$ (the common radius); the points are $A + u_i$ $(i \leqslant 3)$, v, $C + w_j$ $(j \leqslant 2)$, with u_i, v, w_j i.i.d. uniform on S^5. Write the hyperplane as $\delta(x) = \langle n, x\rangle - t = 0$; it meets AC iff $\delta(A)\delta(C) \leqslant 0$.

Hint 2. With $\alpha = -\delta(A)$, $\gamma = -\delta(C)$ the incidences read $\langle n, u_i\rangle = \alpha$, $\langle n, w_j\rangle = \gamma$, $\langle n, v\rangle = \frac{\alpha+\gamma}{2}$, while $\langle n, e\rangle = \frac{\alpha-\gamma}{2}$. Hence the event $\alpha\gamma \leqslant 0$ is exactly $|\langle n, v\rangle| \leqslant |\langle n, e\rangle|$.

Hint 3. The event is invariant under rotating everything, so e may be replaced by a uniform unit vector independent of the rest; now v and e are i.i.d. The substitution $(v, e, w_j) \mapsto (e, v, -w_j)$ preserves the joint law, keeps the same n as a solution of the system (with $\gamma \mapsto -\gamma$), and swaps $|\langle n, v\rangle|$ and $|\langle n, e\rangle|$. The probability is $\frac{1}{2}$.

Problem 7

Hint 1. Four points are not in convex position iff one of them lies in the triangle formed by the other three; these four events are disjoint. Hence $\Pr(\text{convex}) = 1 - 4\,\mathrm{E}|T|/|K|$, where T is the triangle of three random points of K.

Hint 2. The ratio $\mathrm{E}|T|/|K|$ is affine invariant, so for a triangle you may take any convenient one. Compute it (e.g. by Crofton's differential method: let K grow and follow how $|K|^3\mathrm{E}|T|$ changes, or by integrating directly with one vertex conditioned on a side); the result is $\mathrm{E}|T| = |K|/12$, so $\Pr = \frac{2}{3}$.

Hint 3. For the unit disc the same reduction applies; the classical value is $\mathrm{E}|T| = \frac{35}{48\pi^2}|K|$ (integrate in polar coordinates, conditioning on the chord through two of the points), so $\Pr = 1 - \frac{35}{12\pi^2} \approx 0.7045$.

Problem 8

Hint 1. Normalise $\|v_1\| = 1$. The uniform law on Δ can be generated as follows: $s = 1 - p_1$ has density $2s$ on $[0, 1]$, and given s, $(p_2, p_3) = s(q, 1 - q)$ with q uniform. Fix q and put $w = qv_2 + (1 - q)v_3$.

Hint 2. Then $p_1v_1 + p_2v_2 + p_3v_3 = v_1 + s(w - v_1)$ runs along the segment from v_1 to w, at distance sL from v_1, $L = \|w - v_1\|$. The points within distance c of 0 form an interval $\tau_1 \leqslant sL \leqslant \tau_2$ with $\tau_1\tau_2 = 1 - c^2$ (power of v_1) and $\tau_2 - \tau_1 \leqslant 2c$.

Hint 3. The conditional probability is $\int 2s\,ds = \frac{\min(\tau_2, L)^2 - \tau_1^2}{L^2}$, maximal over L at $L = \tau_2$, where it equals $1 - (\tau_1/\tau_2)^2$. Since $\tau_1/\tau_2 \geqslant \frac{1-c}{1+c}$, the bound is $\frac{4c}{(1+c)^2} \leqslant 4c$; integrate over q.

Problem 9

Hint 1. Given $\varepsilon > 0$, choose a convex polygon $P \subset K$ with $|K \setminus P| < \varepsilon$, and around each vertex of P a small set of positive area inside K. If every one of these finitely many sets contains a sample point, the hull nearly contains P. The probability of this tends to 1.

Hint 2. For rates use Efron's identity: $\mathrm{E}|K \setminus C_n| = |K| \Pr(X_{n+1} \notin C_n) = |K| \frac{\mathrm{E}V_{n+1}}{n+1}$, with V the number of hull vertices.

Hint 3. Heuristic: the missed region consists of caps of probability about $1/n$. For a disc such a cap has width $\sim n^{-2/3}$ and $\mathrm{E}V_n \asymp n^{1/3}$, so the deficit is of order $n^{-2/3}$; for a polygon with r vertices the caps sit at the corners and $\mathrm{E}V_n \sim \frac{2r}{3}\ln n$, so the deficit is $\sim \frac{2r}{3}|K|\frac{\ln n}{n}$, much smaller.

Problem 10

Hint 1. If the points lie in some semicircle, exactly one of them (a.s.) is the starting point of a clockwise semicircle containing all the others. Let E_i be the event that the semicircle starting at P_i contains all the points: the E_i are disjoint and $\Pr(E_i) = 2^{1-n}$. So the probability is $n2^{1-n}$.

Hint 2. The analogue: n points uniform on S^{d-1} (or any symmetric, generic law) lie in a common open half-space through the origin with probability $2^{1-n}\sum_{k=0}^{d-1}\binom{n-1}{k}$ (Wendel). Flip signs: each of the 2^n sign patterns $\pm x_i$ is equally likely to work.

Hint 3. The number of sign patterns (ε_i) for which some y has $\varepsilon_i\langle y, x_i\rangle > 0$ for all i equals the number of regions cut out by n generic hyperplanes through 0 in $\mathbb{R}^d$, which is $C(n,d) = 2\sum_{k=0}^{d-1}\binom{n-1}{k}$ (recursion $C(n,d) = C(n-1,d) + C(n-1,d-1)$). Check $d = 2$: $2n/2^n$.